C-HI One-Node Particle Segment Quick-look

Date: 2026-05-09

Summary

This is the first quick-look diagnostic for the completed one-node, particles-enabled `C-HI` segment on `kunanyi`.

Inputs:

- source run: `690426.kunanyi-ohpc`
- quick-look extraction job: `690451.kunanyi-ohpc`
- HDF5 file inspected: `/scratch/bwbrook/xray_cavities_runs/2026-05-08-c-hi-segment/C-HI/particles/problem/outputs/data.0001.flt.h5`
- timestep: `Timestep_1`
- PLUTO code time: `137.169223`
- physical time: `0.4474 Myr`, using `306.601394` PLUTO code units per Myr

This is an early run-health diagnostic, not a science measurement. The aim was to confirm that the jet, tracer, HDF5, and post-processing path behave sensibly before deciding whether to continue, restart, or promote to a larger rehearsal.

Extraction

Kunanyi's login-node Python does not currently expose `h5py`, `numpy`, or `matplotlib`, so the heavy HDF5 read was done with a small serial HDF5 C extractor on a compute node. That job wrote compact TSV slices and profiles which were then plotted locally through the repo `uv` environment.

The reusable tooling added for this is:

- `scripts/science/pluto_h5_quicklook_extract.c`
- `scripts/science/plot_pluto_h5_quicklook.py`
- `scripts/kunanyi/c_hi_quicklook_extract.pbs`

Tracer Extent

The default Level 2 truth threshold remains `tracer > 5e-3`. At this early time the tracer-defined region is compact and symmetric:

Threshold	Cells	x range (kpc)	y range (kpc)	z range (kpc)
1e-3	790808	-2.38 to 2.38	-2.38 to 2.38	-6.32 to 6.32
5e-3	670824	-2.33 to 2.33	-2.33 to 2.33	-6.22 to 6.22
1e-2	619128	-2.28 to 2.28	-2.28 to 2.28	-6.22 to 6.22

Figures

Full-domain mid-plane maps:

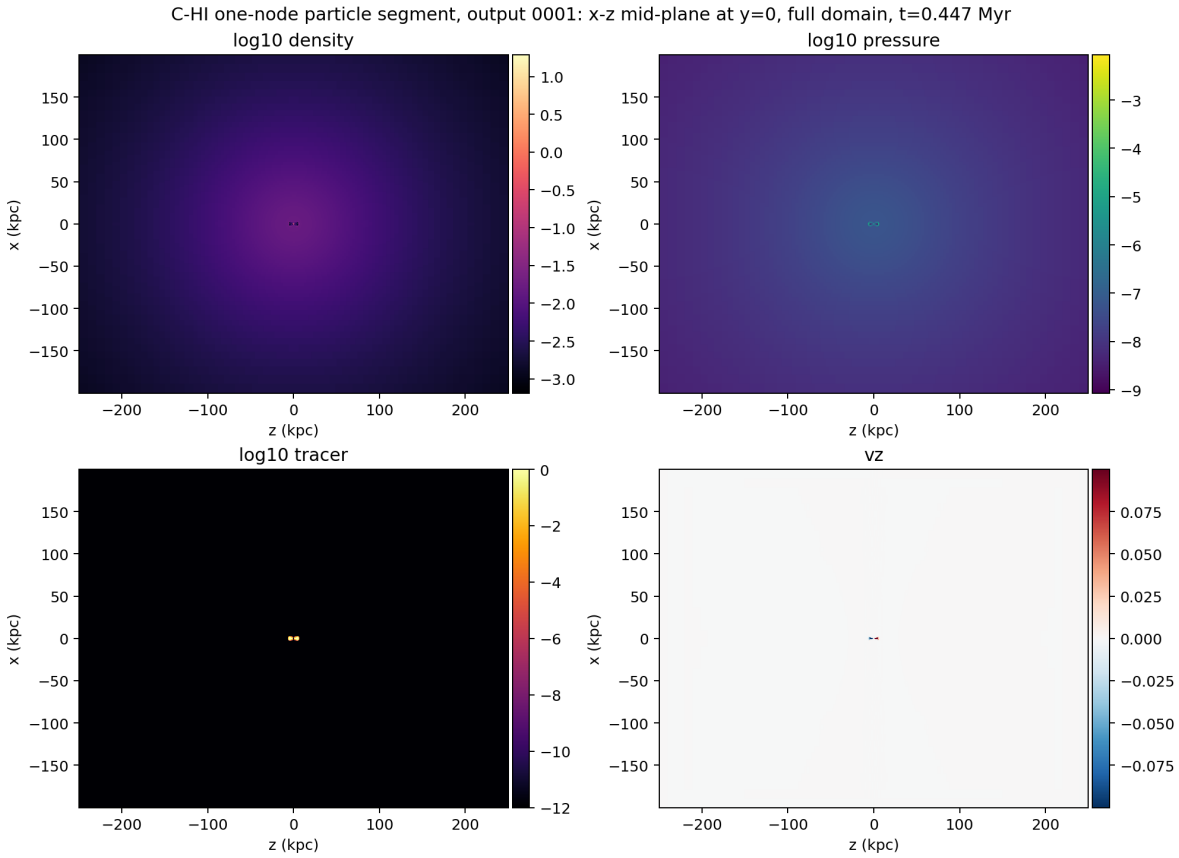


Figure 1: C-HI full-domain quick-look maps

Central zoom:

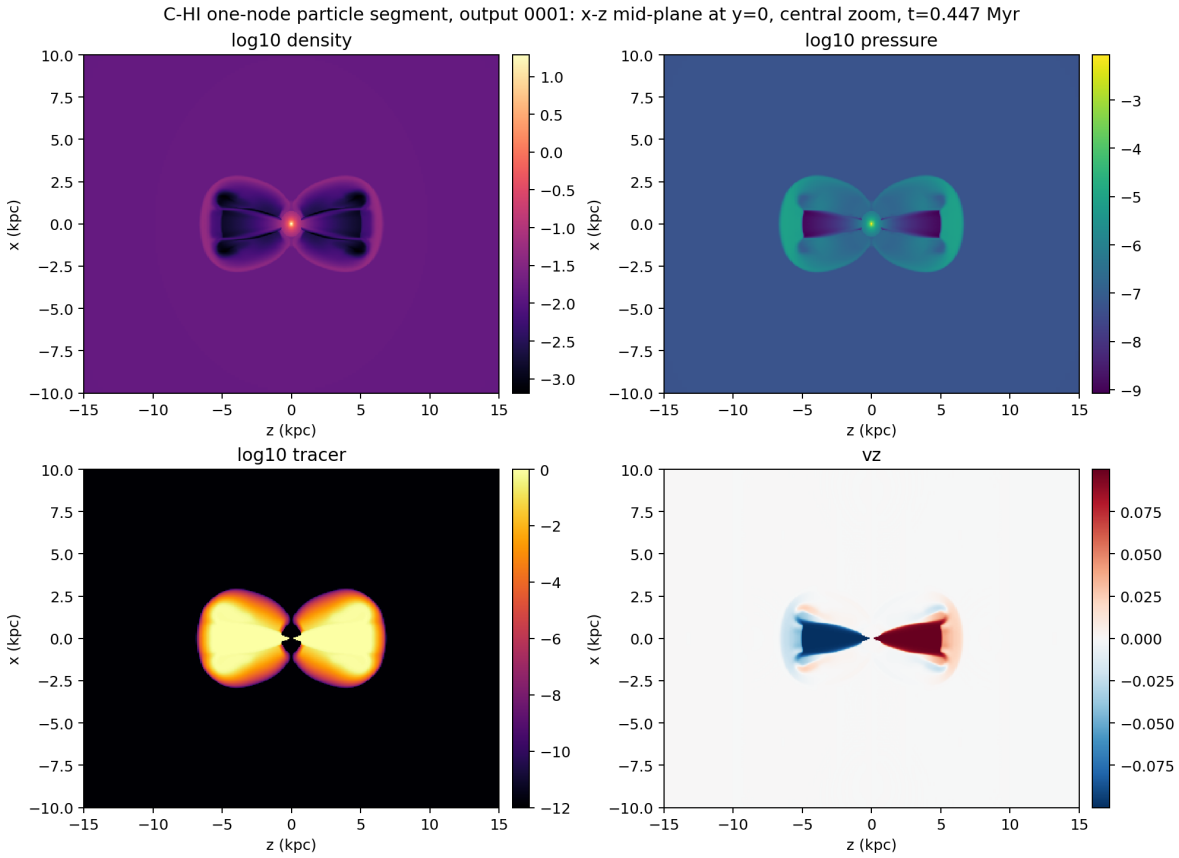


Figure 2: C-HI central quick-look maps

Jet-axis profiles:

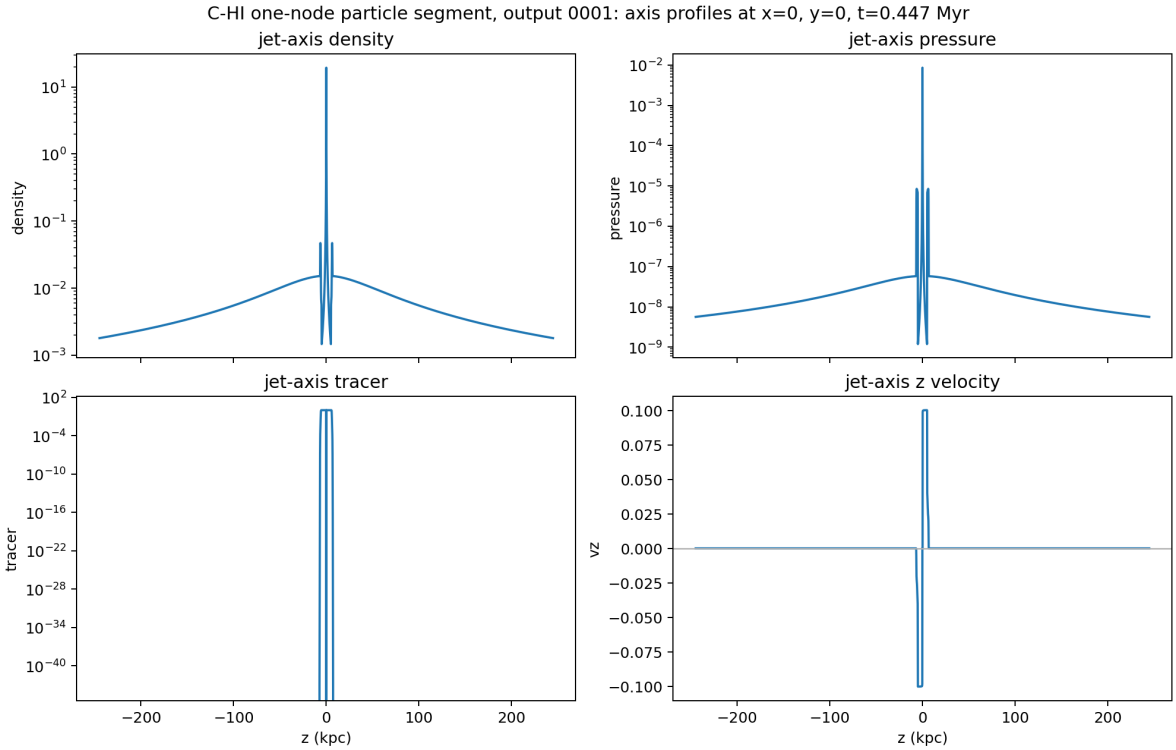


Figure 3: C-HI quick-look axis profiles

Interpretation

The quick-look products pass the immediate sanity gate:

- bipolar tracer lobes are present and symmetric
- the velocity field shows the expected oppositely directed bipolar flow
- density and pressure respond around the early lobes
- the tracer extent is far from the outer grid boundaries
- HDF5 extraction and local plotting are now reproducible without installing a private Python stack on `kunanyi`

The run is still only at `0.4474 Myr`, so it should not be judged for final cavity morphology. The next useful simulation step is a longer restart segment or an intermediate multi-node rehearsal once we decide whether to stay with the one-node shakedown grid or move toward the final supervisor grid.